

Weighted Graph Stats

Weighted Graphs, Edge Lists · Mentor-Led Lab (Student Edition)

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Difficulty ★★ Intermediate

Problem

Read an undirected weighted graph and report its basic figures: the number of vertices and edges, the total edge weight, and the minimum and maximum edge weight.

Implement the edge-list reading and the aggregates from scratch.

What to Compute

- Nodes: the number of vertices N .
- Edges: the number of edges M .
- Total Weight: the sum of all edge weights.
- Min Edge / Max Edge: the smallest and largest edge weight.

Input / Output Format

Input: Line 1: N M . Next M lines: an undirected weighted edge u v w .

Output: Five lines: Nodes, Edges, Total Weight, Min Edge, Max Edge.

Examples

Example 1

Input: 4 5 0 1 1 0 2 3 1 2 1 1 3 4 2 3 2

Output:

Nodes: 4
Edges: 5
Total Weight: 11
Min Edge: 1
Max Edge: 4

Explanation: Five edge weights (1, 3, 1, 4, 2) sum to 11; the cheapest is 1 and the dearest is 4.

Example 2

Input: 3 3 0 1 5 1 2 5 0 2 5

Output:

Nodes: 3
Edges: 3
Total Weight: 15
Min Edge: 5
Max Edge: 5

Explanation: Three equal-weight edges total 15; min and max are both 5.

Constraints

- The graph is undirected and weighted.
- Each edge contributes its weight once to the total.
- Min and max are over individual edge weights.

Sample Run

Sample Input

```
4 5
0 1 1
0 2 3
1 2 1
1 3 4
2 3 2
```

Sample Output

Nodes: 4
Edges: 5
Total Weight: 11
Min Edge: 1
Max Edge: 4